

Proton beam irradiation of uveal melanoma involving the iris, ciliary body and anterior choroid without surgical localisation (light field)

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ABSTRACT

Aims To assess treatment outcomes after proton beam irradiation (PBI) without surgical localisation of uveal melanomas involving the iris, ciliary body and anterior choroid.

Methods Retrospective chart review of 125 patients evaluated at Massachusetts Eye and Ear and treated with PBI using a light field set-up without localisation surgery between November 1975 and April 2017. The tumours were characterised as follows: iris (n=18, 14.4%), ciliary body (n=12, 9.6%), iridociliary (n=58, 46.4%), ciliochoroidal (n=24, 19.2%) and iridociliochoroidal (n=13, 10.4%). The tumours were measured by transillumination and ultrasonography before treatment. Tumours with posterior margin located less than two disc diameters from the ora serrata were treated using the light field technique. Patient outcomes after PBI were evaluated.

Results Most patients had good vision at the time of tumour diagnosis (69.6% had baseline visual acuity (VA) of $\geq 20/40$). Median VA at last follow-up (median follow-up: 72.1 months) was 20/63. Recurrences occurred in 12 patients (9.6%) at a median time of 4.0 years post-treatment. Recurrences were treated by repeat PBI (n=5) or enucleation (n=7). Secondary enucleation was performed in 18 patients (14.4%), and 61.1% of these were due to complications. Neovascular glaucoma (NVG) developed in 21 patients (16.8%). Of seven patients who developed NVG after anti-vascular endothelial growth factor (anti-VEGF) therapies became available, five were treated with intravitreal Avastin injections (23.8% of patients with NVG). Of 69 patients diagnosed with cataract after treatment, 51 (73.9%) were characterised as radiation-related. Death from metastatic uveal melanoma occurred in 20.8% of the cohort, with a median follow-up of 10.1 years.

Conclusions Patients treated with PBI using a light field set-up technique experience good outcomes after irradiation. Eye preservation and retention of good VA are seen in the majority of cases, and tumour recurrence is low.

INTRODUCTION

Proton beam irradiation (PBI) using a light field technique has been shown to be effective in the treatment of circumscribed choroidal haemangioma.¹ Similarly, this technique provides excellent tumour control for choroidal metastases.² Over 90% of patients in our ocular proton therapy programme undergo, prior to irradiation, a surgical procedure

to localise the tumour and suture fiducial markers (tantalum rings), which are then employed in radio-graphical set-up for treatment delivery.^{3,4} Where surgery is not indicated, accurate patient set-up can be performed with a light field technique, in which a visible light field coincident with the proton radiation field on the surface of the eye is used to position the beam relative to the tumour.

Anteriorly located uveal tumours can be treated with PBI without the need for surgical localisation. Tumour margins can be defined using photography (eg, slit lamp photography for iris tumours) and/or transillumination to capture the extent of ciliary body involvement in relation to anatomical landmarks such as the iris and limbus.³⁻⁵ A customised treatment plan is developed with an additional 1 mm margin added to the otherwise standard 3 mm margin between tumour and field edge in fiducial marker-guided treatment to ensure appropriate treatment margins.⁶

Patients with anteriorly located tumours are at lower risk for developing radiation retinopathy in comparison to those with posterior tumours.⁷ However, complications including neovascular glaucoma (NVG) and radiation-induced cataract are more common in this population.⁸

We evaluated our experience of treating patients with uveal melanoma with the light field technique. Treatment outcomes including complications, recurrence and mortality were assessed in all patients in this cohort. We particularly focused on the rate of local tumour control.

METHODS

One hundred and twenty-five patients were evaluated at Massachusetts Eye and Ear (MEE) and Massachusetts General Hospital (MGH) between November 1975 and April 2017 for anterior uveal melanomas and subsequently treated with PBI without surgical placement of fiducial markers. All patients were treated at Harvard Cyclotron (prior to 2002) or the Francis H. Burr Proton Therapy Center at MGH with PBI using a light field set-up technique.

As seen in figure 1A,B, a treatment plan is developed for each tumour based on dimensions obtained by the initial examination and imaging. The prescribed dose, corrected for proton's relative biological effectiveness (RBE), was 70 Gy RBE. There were a few exceptions (n=7), who received between 49.9 and 64.4 Gy. These patients were



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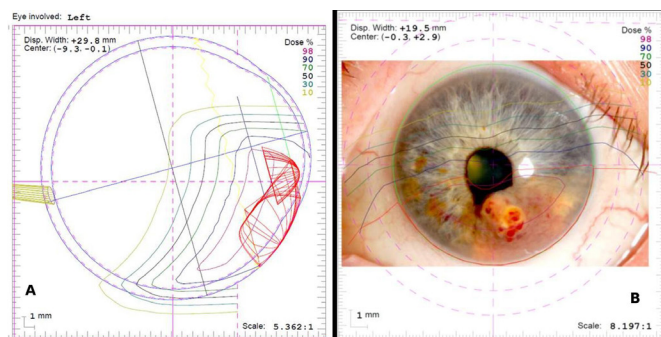


Figure 1 A. Isodose curve generated from proton beam irradiation treatment plan for patient with iris melanoma. B. Isodose curve overlaid with slit lamp photograph.

treated with a lower dose based on the clinical judgement of the treating physician. Based on the tumour characteristics/location of these cases, it was felt that a lower dose would adequately treat the tumours while potentially reducing the risk of radiation complications. The tumours were characterised as follows: iris ($n=18$, 14.4%), ciliary body ($n=12$, 9.6%), iridociliary ($n=58$, 46.4%), ciliochoroidal ($n=24$, 19.2%) and iridociliochoroidal ($n=13$, 10.4%). The tumours were measured by transillumination before treatment. Tumours with posterior margin located less than two disc diameters from the ora serrata were treated using the light field technique.

Post-treatment, patients were re-evaluated at 6 weeks and then every 6 months for 5 years. After 5 years, patients were followed and evaluated on an annual basis for tumour recurrence, complications, eye loss and visual loss.

Mortality status and cause of death for deceased patients in the cohort were obtained through periodic searches of the medical record, hard copy death certificates and the National Death Index (NDI). Some dates and causes of death were obtained through active clinical surveillance.

Cumulative rates of melanoma-specific mortality were calculated using the Kaplan-Meier method. Rates of tumour recurrence, complications, visual loss and secondary enucleations were also estimated based on status at the time of last follow-up. Follow-up for ocular outcomes was completed through March 2019, while mortality follow-up was completed through December 2017, the date through which the NDI was current at the time of our search.

RESULTS

Median age at treatment in the cohort was 61.4 years, and 40.8% ($n=51$) were men. Median largest basal diameter and tumour thickness were 10.0 mm and 4.0 mm, respectively. Median follow-up for ocular outcomes was 6 years (72.1 months), while median follow-up for mortality was 10.1 years.

Most patients had relatively good vision at the time of tumour diagnosis. A total of 87 (69.6%) had a baseline visual acuity (VA) of greater than or equal to 20/40. At the last follow-up visit (median: 6.0 years post-treatment), the median VA was 20/63. A total of 58.4% of the cohort had VA greater than 20/200, while 35.2% had VA greater than 20/40.

NVG developed in 21 patients (16.8% of the cohort); over one-third ($n=8$, 38.1%) of these patients had iridociliary tumours. Three (16.7%) patients with iris-only tumours developed NVG, similar to patients who were diagnosed with ciliochoroidal ($n=4$, 16.7%) and iridociliary ($n=8$, 13.8%) tumours. Patients with iridociliochoroidal tumours were somewhat more

likely to develop NVG ($n=3$, 23.1%). Of the 21 patients who developed NVG, 7 patients developed this complication after anti-vascular endothelial growth factor (anti-VEGF) therapies became available and 5 (23.8%) were treated with intravitreal injections of Avastin (mean number of treatments=3.8 injections). Four patients maintained VA of 20/125 or better after receiving their last injection.

At the time of initial ophthalmological examination, 10 patients had undergone cataract extraction in the affected eye. Of the remaining 115 patients, cataracts were present in 55 patients. Following treatment, cataracts were present in 69 patients (60%), and most cases (73.9%, $n=51$) were considered to be radiation-related. Cataract extraction was performed in 37 of these 69 patients (38%).

Local tumour recurrence occurred in 9.6% ($n=12$) of patients, eight patients with iridociliary tumours, accounting for two-thirds of all recurrences. In contrast, there were no recurrences in the group of patients with ciliochoroidal tumours. One patient (5.6%) with iris melanoma experienced a recurrence, and growth of the tumour was observed in two patients (16.7%) with ciliary body tumours. Most recurrences were ring melanomas ($n=4$) and tumours with heavy pigmentation 360° around the angle ($n=6$). In both types of recurrences, the regrowth occurred away from the initial treatment area. There was one marginal recurrence as well as one unconfirmed recurrence, which was diagnosed and treated outside the MEE. The majority of recurrences ($n=8$) occurred within 5 years of initial treatment with PBI, and the cumulative 5-year recurrence rate was 8.3% (95% CI: 4.2% to 16.1%). Median time from PBI to recurrence was 4.0 years. Patients who experienced recurrences were more likely to develop metastasis than those who remained recurrence-free, 41.7% and 18.6% ($p=0.07$, Fisher's exact test). All recurrences were treated, five with repeat PBI using the light-field technique, and seven required secondary enucleation.

Enucleation was necessary in 18 patients (14.4% of the total cohort); over half ($n=11$, 61.1%) of these were due to complications, including NVG ($n=4$, 22.2%) and other angle closure glaucoma ($n=2$, 11.1%).

The cumulative rates of melanoma-free survival, seen in figure 2, were 86.4%, 79.9% and 77.3% at 5, 10 and 15 years after radiation, respectively. Less than half of the cohort ($n=56$) was still alive at the time of the censor date (31 December 2017). Of the currently living patients, none had

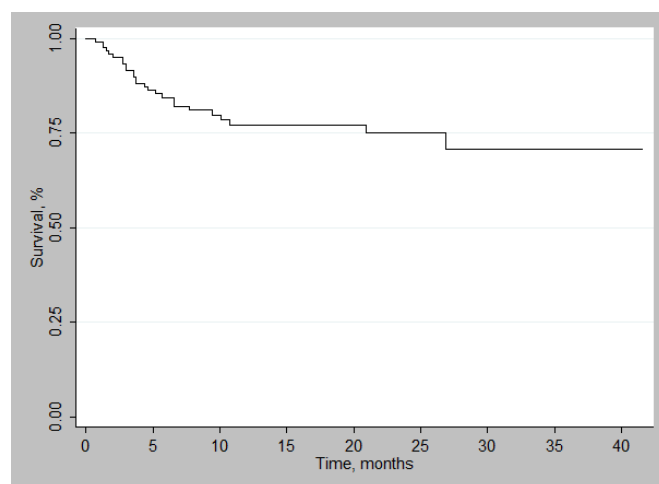


Figure 2 Cumulative probability of metastasis-free survival after proton beam irradiation.

been diagnosed with metastatic disease by this date. Death from metastatic uveal melanoma occurred in 20.8% (n=26) of the cohort; 5 of the 26 patients who died of melanoma (19.2%) had recurrent melanoma and were retreated prior to developing metastasis. The majority of melanoma-related deaths (n=16, 61.5%) occurred within 5 years of diagnosis and treatment, six melanoma deaths occurred between 5 and 10 years after treatment. Only four deaths occurred more than 10 years from diagnosis and treatment; two of these occurred at 21 years and 26.9 years post-treatment. The highest number of melanoma-related deaths occurred in patients with iridociliary (n=11, 42.3%) melanomas, followed by iridociliochoroidal (n=6, 23.1%) melanomas. Four deaths (15.4%) occurred in patients with ciliochoroidal melanomas and ciliary body tumours. One patient (3.8%) diagnosed with an iris-only melanoma developed metastasis in this cohort.

DISCUSSION

PBI using a light field technique is an attractive modality to treat anterior uveal melanoma, as surgery can be avoided. There is a theoretical risk that tumour localisation could be less precise than otherwise obtained with surgical localisation, resulting in incomplete coverage of the tumour and leading to increased risk of recurrence. Based on the experiences of this cohort, our findings suggest that anteriorly located tumours can be adequately controlled with PBI using a non-surgical, light field technique for treatment set-up. It was not feasible to complete a formal analysis evaluating the light field technique as a risk factor for tumour recurrence. However, the patterns of recurrence we observed in this cohort support the hypothesis that using this technique does not negatively affect local tumour control. Most recurrences (more than 80%) arose in cases of ring melanomas or tumours with heavy pigmentation in the entire angle, and the regrowth occurred away from the initial treatment area. Melanomas with ciliary body involvement are known to be at higher risk of recurrence.⁹ In the case of ring melanomas in particular, it can be difficult to visualise these tumours due to their occult nature.^{10 11} Similarly, it is difficult to determine malignant cells from macrophages in angles with heavily pigmented cells, and the increased risk of radiation complications must be weighed against the risk of incomplete coverage of the tumour when identifying the extent of tumour and area to treat.

Although direct comparisons are difficult to make due to intrinsic differences between these tumours and more posteriorly located tumours that require surgical localisation before PBI, rates of recurrence in this cohort are higher than rates reported for choroidal and ciliary body tumours treated with proton therapy.^{9 12–15} In two large studies, the 5-year rate of recurrence was 4%,^{12 13} and in another study, 3.6% of patients experienced a recurrence¹⁵ during the study period. In a previous study completed at MEE/MGH,⁹ the 5-year rate of local failure was 3.2% and increased to 4.3% at 10 years. Riechardt *et al* reported a higher recurrence rate of 6.1%¹⁴ in 147 patients with parapapillary tumours treated with PBI.

Investigators at the Douglas Cyclotron, Clatterbridge Centre for Oncology have treated iris melanomas without surgical localisation. In one series of patients, Damato and colleagues found that 6.3% recurred at 5 years after treatment,¹⁶ and a more recent report of patients treated at this facility, stated that 5.3% of patients had experienced tumour regrowth.¹⁷ At the Centre Lacassagne, Nice where fiducial markers were

placed in 107 patients with iris melanomas before PBI was completed, the 5-year recurrence rate was 7.5%.¹⁸

In all of these studies, patients with ciliary body tumours accounted for a small percentage of the total cohort while the vast majority of patients in this study had tumours that involved the ciliary body. Given that ciliary body involvement is a known risk factor for tumour recurrence,^{9 19} this difference in distribution of tumour types may be one explanation for the difference in rates.

Rates of metastasis or melanoma-related deaths, on the other hand, were similar between patients in this study and other studies of PBI. We found that 20.8% of the patients in this cohort died of metastatic melanoma, virtually identical to that reported in earlier studies of choroidal/ciliary body tumours treated with PBI,^{13 19} and a more recent study of 508 patients treated with PBI over a 20-year period.²⁰

Anterior tumours treated with brachytherapy appear to have similar functional outcomes to those treated with proton radiation. Gunduz *et al*²¹ evaluated 136 patients who had completed I-125 treatment for uveal melanomas with predominantly ciliary body involvement and found that 21% developed NVG. A somewhat lower rate, 15%, was observed in another study of patients treated with I-125 for tumours that involved the ciliary body.²² We found that 16.8% of patients in the current study developed NVG after PBI. The enucleation rate also compared favourably with that reported after I-125 radiotherapy, 14.4% for PBI-treated patients and 15%²¹ and 7.1% of I-125-treated patients.²² In contrast, Finger reported no cases of NVG and no cases of eye removal in 22 patients treated with Pd-103, although radiation-related cataracts occurred in a larger number of patients, 71%,²³ than those reported after I-125 and PBI.

Among these patients treated with brachytherapy, regrowth occurred in 2.4% in one study²² and 8% of patients in another.²¹ These differences in recurrences may be attributable to differences in follow-up time (43 months vs 70 months). Of note, follow-up time and recurrence rate in our cohort was similar to that reported by Gunduz *et al*.²¹

Our study findings demonstrate that reasonable rates of local control are achieved with PBI using a light field technique. The rate of melanoma-related mortality is comparable to rates seen in other cohorts. Rates of eye-related complications are also similar to that achieved with brachytherapy.

PBI using a light field technique can be an appropriate treatment to manage anterior uveal melanoma. Eye retention and good VA are seen in the majority of cases, and tumour recurrence is low.

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Patient consent for publication Not required.

Ethics approval This retrospective chart review was performed with approval from the MEE Institutional Review Board.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available upon reasonable request. Data are stored as deidentified information in the Uveal Melanoma Repository at Mass Eye and Ear. Any requests to access deidentified information would need to be approved by the Institutional Review Board that guides our research.

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REFERENCES

- 1 Chan RVP, Yonekawa Y, Lane AM, *et al.* Proton beam irradiation using a light-field technique for the treatment of choroidal hemangiomas. *Ophthalmologica* 2010;224:209–16.
- 2 Tsina EK, Lane AM, Zacks DN, *et al.* Treatment of metastatic tumors of the choroid with proton beam irradiation. *Ophthalmology* 2005;112:337–43.
- 3 Gragoudas ES. Proton beam irradiation of uveal melanomas: the first 30 years. The Weisenfeld lecture. *Invest Ophthalmol Vis Sci* 2006;47:4666–73.
- 4 Lu JE, Welch RJ, Mishra KK, *et al.* Ultrasonography and transillumination for uveal melanoma localisation in proton beam treatment planning. *Eye* 2019;33:1904–10.
- 5 Kamran SC, Collier JM, Lane AM, *et al.* Outcomes of proton therapy for the treatment of uveal metastases. *Int J Radiat Oncol Biol Phys* 2014;90:1044–50.
- 6 Goitein M, Miller T. Planning proton therapy of the eye. *Med Phys* 1983;10:275–83.
- 7 Riechardt AI, Pilger D, Cordini D, *et al.* Neovascular glaucoma after proton beam therapy of choroidal melanoma: incidence and risk factors. *Graefes Arch Clin Exp Ophthalmol* 2017;255:2263–9.
- 8 Finger PT. Tumour location affects the incidence of cataract and retinopathy after ophthalmic plaque radiation therapy. *Br J Ophthalmol* 2000;84:1068–70.
- 9 Gragoudas ES, Lane AM, Munzenrider J, *et al.* Long-Term risk of local failure after proton therapy for choroidal/ciliary body melanoma. *Trans Am Ophthalmol Soc* 2002;100:43–8. discussion 8–9.
- 10 Simpson ER. Ciliary body melanoma: a special challenge. *Can J Ophthalmol* 2004;39:365–71.
- 11 Vásquez LM, Pavlin CJ, McGowan H, *et al.* Ring melanoma of the ciliary body: clinical and ultrasound biomicroscopic characteristics. *Can J Ophthalmol* 2008;43:229–33.
- 12 Egger E, Schalenbourg A, Zografos L, *et al.* Maximizing local tumor control and survival after proton beam radiotherapy of uveal melanoma. *Int J Radiat Oncol Biol Phys* 2001;51:138–47.
- 13 Dendale R, Lumbroso-Le Rouic L, Noel G, *et al.* Proton beam radiotherapy for uveal melanoma: results of Curie Institut–Orsay proton therapy center (ICPO). *Int J Radiat Oncol Biol Phys* 2006;65:780–7.
- 14 Riechardt AI, Cordini D, Willerding GD, *et al.* Proton beam therapy of parapapillary choroidal melanoma. *Am J Ophthalmol* 2014;157:1258–65.
- 15 Seibel I, Cordini D, Rehak M, *et al.* Local recurrence after primary proton beam therapy in uveal melanoma: risk factors, retreatment approaches, and outcome. *Am J Ophthalmol* 2015;160:628–36.
- 16 Damato B, Kacperek A, Chopra M, *et al.* Proton beam radiotherapy of iris melanoma. *Int J Radiat Oncol Biol Phys* 2005;63:109–15.
- 17 Sandinha MT, Kacperek A, Errington RD, *et al.* Recurrence of iris melanoma after proton beam therapy. *Br J Ophthalmol* 2014;98:484–7.
- 18 Thariat J, Rahmi A, Salleron J, *et al.* Proton beam therapy for iris melanomas in 107 patients. *Ophthalmology* 2018;125:606–14.
- 19 Gragoudas E, Li W, Goitein M, *et al.* Evidence-Based estimates of outcome in patients irradiated for intraocular melanoma. *Arch Ophthalmol* 2002;120:1665–71.
- 20 Bellocq D, Roy P, Kodjikian L, *et al.* 20-Year assessment of metastatic latency and subsequent time to death after proton therapy for uveal melanomas. *Melanoma Res* 2020;30:272–8.
- 21 Gündüz K, Shields CL, Shields JA, *et al.* Plaque radiotherapy of uveal melanoma with predominant ciliary body involvement. *Arch Ophthalmol* 1999;117:170–7.
- 22 Krema H, Simpson ER, Pavlin CJ, *et al.* Management of ciliary body melanoma with iodine-125 plaque brachytherapy. *Can J Ophthalmol* 2009;44:395–400.
- 23 Finger PT. Plaque radiation therapy for malignant melanoma of the iris and ciliary body. *Am J Ophthalmol* 2001;132:328–35.